Project Title: Machine Learning Framework for Skill Gap Analysis and Personalized Learning Recommendation

Abstract

The rapid evolution of technology has significantly increased the demand for job-specific skills across various industries. However, many students and professionals face challenges in identifying the gap between their existing skills and the competencies required for their desired career paths. This project, **"Machine Learning Framework for Skill Gap Analysis and Personalized Learning Recommendation,"** aims to develop an intelligent system that automatically analyzes a user's current skill set, identifies missing competencies for a selected job role, and recommends a personalized learning path to bridge those gaps.

The proposed framework utilizes Natural Language Processing (NLP) and machine learning techniques to compare user skills with job descriptions collected from industry datasets. Using sentence embeddings and semantic similarity, the system understands the contextual relationship between skills rather than relying solely on keyword matching. Based on the identified skill gaps, the framework recommends relevant courses, certifications, and learning resources in an optimized sequence, enabling users to efficiently prepare for their target careers.

The system consists of modules for resume or profile analysis, job role selection, skill extraction, semantic skill matching, skill gap identification, and personalized learning recommendation. Machine learning models and semantic similarity algorithms are employed to improve recommendation accuracy and provide meaningful insights into the user's strengths and areas for improvement.

This framework assists students, job seekers, and working professionals in making informed learning decisions, reducing the time required to identify appropriate learning resources, and enhancing career readiness. By integrating machine learning with intelligent recommendation techniques, the proposed system provides an effective solution for personalized skill development in today's competitive job market.

Datasets:

Dataset	Purpose	URL Link
O*NET Skills Database	Provides skills required for different job roles	https://www.onetcenter.org/database.html
Kaggle Job Skills Dataset	Job descriptions and required technical skills	https://www.kaggle.com/
Coursera/Udemy Course Dataset	Learning resources for missing skills	Kaggle or custom dataset
Custom Student Skills Dataset	Student's current skills, proficiency, interests and career goal	Self-created using Google Forms or Excel

Proposed Approach:



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